Interior-zero complexity of polynomial differentiation residuals

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Abstract

We classify the numbers of interior zeros that permit vanishing and superalgebraic differentiation residuals for polynomials whose zeros lie in the closed unit disk. If a degree-n polynomial has exactly M strictly interior zeros, then, for every complex slope, its normalized H^2 residual is at least $n/[2\sqrt{2}\,16^M(n+2M+4)^{2M+2}]$. Consequently a prescribed budget b_n permits a superalgebraically small residual sequence if and only if $b_n \to \infty$. Sufficiency holds with real coefficients, normalization at the origin, the slope fixed by that normalization, and exactly b_n interior zeros. In contrast, one interior zero already permits residuals $O(n^{-1/6})$, with family-specific asymptotic constant $\sqrt{15}/4^{4/3}$, whereas the circle-only infimum tends to $1/\sqrt{8}$. For one interior zero we also prove a universal $cn^{-13/6}$ lower bound; the optimal rate remains open. The finite-budget obstruction combines a reciprocal identity with classical Mahler, truncated-shift and Poisson-kernel estimates, and extends to a stable factor times a self-inversive cofactor. For unrestricted interior counts we prove $\exp(-C\sqrt{n})$ lower bounds for every $C > \sqrt{(8/3)\log 17}$. The upper construction corrects both the exact slope and degree rounding at every large degree. The resulting refinement of exterior power-sum bounds uses a known residual method; the negative answer to Erdős Problem 973 was already known. No matching fixed-budget rates or stretched-exponential upper bounds are claimed.

1 Statement and relation to previous work

All logarithms are natural. The measure on the unit circle is normalized to have mass one. Thus for $P(z) = \sum_{j=0}^{n} a_j z^j$,

$$\|P\|_2^2 = \frac{1}{2\pi}\int_0^{2\pi} |P(\mathrm{e}^{it})|^2\,dt = \sum_{j=0}^{n} |a_j|^2, \qquad \|P\|_\infty = \max_{|z|=1} |P(z)|.$$

Write $\mathcal{S}_n$ for the nonzero polynomials of exact degree n whose zeros are in $\overline{\mathbb{D}}$, including multiplicity and boundary zeros. Define

$$s_n = \inf_{\substack{P\in\mathcal{S}_n \\ \lambda\in\mathbb{C}}} \frac{\|P' - \lambda P\|_2}{\|P\|_2}.$$

There is no normalization at the origin in this definition.

For an individual nonzero polynomial write $\rho_\lambda(P) = \|P' - \lambda P\|_2/\|P\|_2$. Our principal classification concerns the number of strictly interior zeros; all root counts include multiplicity.

Theorem 1 (Finite interior-root budget)**.** *Let $P \in \mathcal{S}_n$ have exactly M zeros in $\mathbb{D}$, where $n \geq 1$ and $0 \leq M \leq n$. For every $\lambda \in \mathbb{C}$,*

$$\rho_\lambda(P) \geq \frac{n}{2\sqrt{2}\,16^M(n+2M+4)^{2M+2}}. \tag{1}$$

In particular, for $n \geq 2M+4$, $\rho_\lambda(P) \geq 2^{-6M-4}n^{-2M-1}$. Neither the exponent nor the constant is asserted to be sharp.

Theorem 2 (Exact criterion for a growing budget)**.** *Let (b_n) be any integer sequence with $0 \le b_n \le n$. The following are equivalent:*

1. *$b_n \to \infty$.*
2. *There exist $P_n \in \mathcal{S}_n$, with at most b_n strictly interior zeros, and complex slopes λ_n, such that $\rho_{\lambda_n}(P_n) = o(n^{-A})$ for every fixed $A > 0$.*
3. *There exist real-coefficient $P_n \in \mathcal{S}_n$, with $P_n(0) = 1$ and exactly b_n strictly interior zeros, such that $\rho_{P_n'(0)}(P_n) = o(n^{-A})$ for every fixed $A > 0$.*

For residuals merely tending to zero under the "at most" budget, the necessary and sufficient condition is instead $b_n \ge 1$ eventually, with the same choice of restricted slope available for sufficiency.

No monotonicity of the budget is assumed. The sufficiency argument is existential: no explicit quantitative thresholds for an arbitrary budget are supplied, and extra interior zeros may lie arbitrarily close to the circle. Section 7 proves these statements and gives a necessary count for stretched-exponential decay. Their ingredients include classical multiplication and restricted Jensen estimates [8, 9, 10], and the logarithmic derivative of a finite Blaschke product [11, Lemma 4.3]. These tools themselves are not new.

Theorem 3. *Put $C_0 = \sqrt{(8/3)\log 17}$. For every $C > C_0$ there is N_C such that*

$$s_n \ge \mathrm{e}^{-C\sqrt{n}} \qquad (n \ge N_C). \tag{2}$$

The following explicit bound also holds:

$$s_n \ge \mathrm{e}^{-3\sqrt{n}} \qquad (n \ge 10^4). \tag{3}$$

For comparison, Tan, Wang, Huang and Chen [1, Theorem 1.1] considered the restricted infimum

$$r_n = \inf_{\substack{P \in \mathcal{S}_n \\ P(0)=1}} \frac{\|P' - P'(0)P\|_2}{\|P\|_2}$$

and proved $r_n \ge \exp(-(1+o(1))\sqrt{n}\log n)$. For an individual normalized polynomial we write $\rho(P) = \|P' - P'(0)P\|_2/\|P\|_2$. Theorem 5 identifies the positive limiting residual when every root lies on the circle. Theorem 13 shows that one interior root suffices to make it vanish, establishing the minimal zero-count threshold. These statements concern the global weighted boundary norm, including the exact restricted slope.

Since $s_n \le r_n$, Theorem 3 strengthens that quantitative estimate and allows an arbitrary scalar slope. Their maximum-point normalization, segment integration and projected Newton identity are retained here, with attribution. The two changes are a local analytic bound for the logarithmic derivative and a bounded test polynomial. These remove the logarithmic factor in the exponent.

Luo, Yang and Zhu [2, Theorem 1.1] had already proved that exterior power sums cannot be exponentially small in the degree, answering Erdős Problem 973 negatively. The application in Section 5 is a quantitative refinement of that established conclusion. No new solution of the original yes-or-no problem is asserted.

Theorem 4 (Superalgebraic decay)**.** *For every $A > 0$, there are constants C_A, N_A such that*

$$r_n \le C_A n^{-A} \qquad \text{for every integer } n \ge N_A.$$

The polynomials can be chosen with real coefficients, exactly M_A zeros strictly inside the unit disk, where M_A is independent of n, and all other zeros on its boundary. The interior zeros approach 1 with distance and radial gap $\Theta(n^{-1/3})$. Together with Theorem 3, this implies superalgebraic decay of both s_n, r_n, while $s_n^{1/n}, r_n^{1/n} \to 1$.

The proof in Section 6 fixes the cancellation order and finite integer multiplicities first, then lets the reciprocal root scale grow as $n^{1/3}$. No explicit quantitative bounds are supplied here on M_A, C_A, N_A or uniform control as A grows. Section 8 explains why a fixed strict root radius precludes superalgebraic decay of a single sequence.

Approximation by logarithmic derivatives and finite moment realization have substantial prior history. Komarov [3] obtains geometric approximation on each fixed subdisk by logarithmic derivatives of circle-root polynomials; [4] also realizes prescribed initial moments with unit weights. Those interior estimates do not supply the global weighted boundary norm used here, with a target slope growing with degree. For example, $1 + z^n$ approximates zero logarithmically on subdisks but has restricted residual $n/\sqrt{2}$. The present contribution is the combination of exact slope, global concentration and correction at every large degree, rather than moment realization alone.

2 The sharp barrier when every zero is on the circle

Let s_n° and r_n° denote the same infima as s_n, r_n, with every zero required to lie on $|z| = 1$.

Theorem 5. *The unrestricted and restricted circle-root residuals satisfy*

$$\lim_{n\to\infty} s_n^\circ = \lim_{n\to\infty} r_n^\circ = \frac{1}{\sqrt{8}}.$$

The lower asymptotic bound also holds in the larger class of self-inversive polynomials. No finite-degree extremizer is asserted.

Proof. We first prove the lower bound, allowing an arbitrary complex slope. Rotating the variable makes this slope real and nonnegative, denoted λ. Self-inversivity permits a constant phase choice such that

$$P(\mathrm{e}^{it}) = \mathrm{e}^{int/2}p(t), \qquad p(t) \in \mathbb{R}, \qquad p(t + 2\pi) = (-1)^n p(t).$$

These operations preserve the unrestricted normalized residual. The familiar phase derivative $n/2$ for circle-root polynomials appears explicitly in [5, (1.3)–(1.4)]; here we retain the additional scalar subtraction. Direct differentiation gives

$$P' - \lambda P = \mathrm{e}^{i(n/2-1)t}\big[(n/2 - \lambda\cos t)p - i(p' + \lambda \sin t\, p)\big].$$

Integration of the cross term by parts is legitimate because p^2 is periodic. Consequently the squared residual is exactly

$$\frac{\int_0^{2\pi}[p'^2 + (n^2/4 + \lambda^2 - (n+1)\lambda\cos t)p^2]\,dt}{\int_0^{2\pi} p^2\,dt}. \tag{4}$$

Write $E(L)$ for the lowest periodic eigenvalue of $-d^2/dt^2 - L\cos t$, for $L \geq 0$. In odd degree, $|p|$ is periodic in H^1 and has the same quadratic-form values as p. Thus in both parities (4) is at least

$$g_n(\lambda) = n^2/4 + \lambda^2 + E((n+1)\lambda).$$

This is a variational lower bound, not an identification of the periodic and antiperiodic spectra.

The change of variables $t = 2x + \pi$ gives $E(L) = a_0(2L)/4$, in the standard Mathieu convention $-d^2/dx^2 + 2q\cos(2x)$; see [7, §28.2]. The use of this ground state in circular variational problems is classical; compare [6, §VI.A.2]. The fixed-index expansion [7, (28.8.1), $m = 0, s = 1$] is

$$a_0(q) = -2q + 2\sqrt{q} - \tfrac{1}{4} + O(q^{-1/2}), \qquad E(L) = -L + \sqrt{L/2} - \tfrac{1}{16} + O(L^{-1/2}).$$

We must minimize uniformly over λ, rather than applying this expansion only at a fixed shift. The variational principle gives $E(L) \geq -L$ and Lipschitz continuity, so g_n is continuous and coercive, with

$$g_n(\lambda) \geq (\lambda - (n+1)/2)^2 - (2n+1)/4.$$

The trial value $\lambda = n/2 + 1/4$ gives $g_n = 1/8 + O(n^{-1})$. Every minimizer therefore has the form $n/2 + c_n$, where $c_n = O(\sqrt{n})$. On this window $(n+1)\lambda \asymp n^2$, and the displayed single-variable Mathieu remainder is uniformly $O(n^{-1})$. Expanding its square root uniformly on the same window yields

$$g_n(n/2 + c) = (c - 1/4)^2 + \tfrac{1}{8} + O((c^2+1)/n).$$

The upper bound on the minimum and absorption of the c_n^2/n term imply $c_n = O(1)$. Applying the expansion again proves $\inf_{\lambda \geq 0} g_n(\lambda) = 1/8 + O(n^{-1})$. This supplies the required lower limit, also for the restricted slope.

For the upper bound, write $n = 6m + k$, $0 \leq k \leq 5$, and for $m \geq 3$ choose

$$\begin{aligned} c_{m,k} = \cos t_{m,k} &= \tfrac{1}{2} + \frac{2k-1}{8m}, \\ B_t(z) &= (1+z)^4(1 - 2\cos t\, z + z^2), \\ P_{m,k}(z) &= B_{t_{m,k}}(z)^m (1+z)^k. \end{aligned}$$

All zeros lie on the circle, $P_{m,k}(0) = 1$, and

$$P'_{m,k}(0) = n/2 + 1/4, \qquad P'_{m,k}(1)/P_{m,k}(1) = n/2.$$

The limiting factor is $B_* = (1+z)^4(1 - z + z^2)$. With $s = \sin^2(\theta/2)$, its squared normalized modulus is $\cos^6(\theta/2)\cos^2(3\theta/2)$, with a unique maximum at $\theta = 0$, and its logarithm is $-3\theta^2 + O(\theta^4)$. More explicitly, for the perturbed factor this modulus is

$$H_c(s) = (1-s)^4 \left(1 - \frac{2s}{1-c}\right)^2.$$

Its secondary maximum has value $(1+c)^6/[729(1-c)^2] < 1$ for $0 \leq c \leq 3/4$. All six parameter sequences eventually lie in this interval. Local negative curvature and the strict outer gap are uniform as $c \to 1/2$. They give Gaussian domination near zero and an exponentially small outer tail.

For $F_{m,k} = P'_{m,k}/P_{m,k} - P'_{m,k}(0)$, the identity

$$B'_*/B_* - 3 = -\frac{3z(z-1)^2}{z^3+1}$$

shows, on putting $\theta = u/\sqrt{m}$, that

$$F_{m,k}(\mathrm{e}^{i\theta}) \longrightarrow -\tfrac{1}{4} + \tfrac{3}{2}u^2, \qquad |P_{m,k}(\mathrm{e}^{i\theta})/P_{m,k}(1)|^2 \longrightarrow \mathrm{e}^{-3u^2}.$$

Indeed the constant term of $F_{m,k}$ at 1 is exactly $-1/4$, its linear term is $O(1)$, and its quadratic term is $-3m/2 + O(1)$. A fixed zero-free neighborhood gives the local bound $C(1 + |\theta| + m\theta^2)$; after rescaling, a polynomial times a Gaussian dominates. On the outer interval, differentiate the polynomial product, removing one factor. Uniformly bounded normalized factor derivatives give an outer squared numerator $O(m^2 q^{2m-2})$, for some $q < 1$. Thus no estimate at a pole of the logarithmic derivative is required. The ratio of the complete integrals tends to

$$\frac{\int_{\mathbb{R}} (-1/4 + 3u^2/2)^2 \mathrm{e}^{-3u^2}\, du}{\int_{\mathbb{R}} \mathrm{e}^{-3u^2}\, du} = \tfrac{1}{16} - \tfrac{3}{4}\cdot\tfrac{1}{6} + \tfrac{9}{4}\cdot\tfrac{1}{12} = \tfrac{1}{8}.$$

Uniformity over the six values of k covers every sufficiently large degree. This proves the upper bound for r_n°, and $s_n^\circ \leq r_n^\circ$ completes both limits. □

3 A finite residual certificate

We first give conditions that can be checked at a fixed degree. The asymptotic and explicit choices will then be elementary.

Lemma 6 (Normalization and separation)**.** *Suppose $P \in \mathcal{S}_n$, $\lambda \in \mathbb{C}$, $y > 0$, and*

$$\frac{\|P' - \lambda P\|_2}{\|P\|_2} \le \mathrm{e}^{-y}, \qquad \sqrt{n+1}\,\mathrm{e}^{-y} \le 1.$$

Set

$$\gamma_n = 2 + \log(2\sqrt{n+1}), \qquad \eta = \frac{y - \gamma_n}{n}. \tag{5}$$

If $\eta > 0$, there are numbers $v_1, \dots, v_n$ and moments $m_k = n^{-1}\sum_j v_j^k$ such that

$$m_1 \in \mathbb{R}, \quad m_1 \ge \tfrac{1}{2}, \qquad \operatorname{Re} v_j \ge \tfrac{1}{2}, \quad |v_j|^2 \le \frac{\operatorname{Re} v_j}{\eta}. \tag{6}$$

They are the reciprocals $v_j = (1-\beta_j)^{-1}$ of the complex displacements from the zeros to a normalized maximum point.

Proof. This is the normalization and separation argument of [1, Sections 2.1–2.2], with the scalar slope left arbitrary. Choose $|\tau| = 1$ at a maximum of $|P|$, and put

$$G(w) = \frac{P(\tau w)}{P(\tau)}, \qquad \mu = \tau\lambda, \qquad E = G' - \mu G.$$

The denominator is nonzero because P is nonzero. We have $G(1) = \|G\|_\infty = 1$, $\|G\|_2 \le 1$, and $\|E\|_2 \le \mathrm{e}^{-y}$. Differentiating $|G(\mathrm{e}^{it})|^2$ at its maximum shows that $d = G'(1)$ is real. Bernstein's polynomial inequality gives $|d| \le n$. The coefficient bound $\|E\|_\infty \le \sqrt{n+1}\|E\|_2$ therefore yields

$$|\mu - d| = |E(1)| \le 1, \qquad |\mu| \le n + 1. \tag{7}$$

Let β be any zero of G, and set $\delta = 1 - \operatorname{Re}\beta$. Integrating the exact ODE along the segment from β to 1 gives

$$1 = \int_\beta^1 \mathrm{e}^{\mu(1-w)} E(w)\, dw.$$

That segment lies in $\overline{\mathbb{D}}$. On it, $\operatorname{Re}(\mu(1-w)) \le n\delta + 2$, using $d \in \mathbb{R}$, $|d| \le n$, $|\mu - d| \le 1$, and $|1 - \beta| \le 2$. Consequently

$$1 \le 2\sqrt{n+1}\,\mathrm{e}^{n\delta + 2 - y}, \qquad \delta \ge \eta.$$

No zero equals 1, since $G(1) = 1$. For $v = (1-\beta)^{-1}$,

$$\operatorname{Re} v = \frac{1 - \operatorname{Re}\beta}{|1-\beta|^2} \ge \tfrac{1}{2}, \qquad \frac{|v|^2}{\operatorname{Re} v} = \frac{1}{1 - \operatorname{Re}\beta} \le \frac{1}{\eta}.$$

Finally $nm_1 = G'(1)/G(1) = d$, so m_1 is real and at least $1/2$. □

Lemma 7 (Local analytic estimate)**.** *With the normalization above, let $r > 0$ and suppose*

$$q = r\sqrt{n+1}\,\exp\bigl(-y + (2n+1)r\bigr) \le \tfrac{1}{2}. \tag{8}$$

Then G has no zero in $|w - 1| \le r$, and, for every integer $k \ge 2$,

$$|m_k| \le \frac{2\sqrt{n+1}}{n}\exp\bigl(-y + (2n+1)r\bigr) r^{1-k}. \tag{9}$$

Proof. The disk centered at 1 can extend beyond $\overline{\mathbb{D}}$; the needed control there comes from the coefficients. For $|u| \le r$,

$$|E(1+u)| \le \sqrt{n+1}(1+r)^n \|E\|_2 \le \sqrt{n+1}\,\mathrm{e}^{-y+nr}.$$

Solving the ODE from $G(1) = 1$ gives

$$G(1+u) = \mathrm{e}^{\mu u}(1+h(u)), \qquad h(u) = \int_0^u \mathrm{e}^{-\mu s} E(1+s)\,ds.$$

Using the straight segment and (7), we obtain $|h(u)| \le q \le 1/2$. Thus G is nonzero on that closed disk and on a neighborhood of it. The analytic function $F = G'/G - \mu = E/G$ satisfies

$$|F(1+u)| \le 2\sqrt{n+1}\,\mathrm{e}^{-y+(2n+1)r} \qquad (|u| \le r).$$

Cauchy's derivative estimate and logarithmic differentiation give

$$\frac{|F^{(k-1)}(1)|}{(k-1)!} \le 2\sqrt{n+1}\,\mathrm{e}^{-y+(2n+1)r} r^{1-k}, \qquad F^{(k-1)}(1) = (-1)^{k-1}(k-1)!\,nm_k.$$

This proves (9). There is no restriction $k \le n+1$: F is an analytic rational function on the specified disk. □

Proposition 8 (Finite certificate)**.** *Fix $n \ge 1$, $y, r > 0$, and an integer $K \ge 2$. Define η by (5) and set $a = 2\eta$. Suppose $0 < \eta \le 2$, $\sqrt{n+1}\mathrm{e}^{-y} \le 1$, and (8) holds. Put*

$$B = \frac{2\sqrt{n+1}\,r}{Kan} \exp\left(-y + (2n+1)r + K\log\left(1 + \frac{a}{r}\right)\right). \tag{10}$$

If

$$\frac{1}{K\eta} + B < \frac{1}{2}, \tag{11}$$

then $s_n \ge \mathrm{e}^{-y}$.

Proof. Suppose an admissible P, λ had residual at most e^{-y}. For the numbers in Lemma 6,

$$|1 - av_j|^2 = 1 - 2a\,\mathrm{Re}\,v_j + a^2|v_j|^2 \le 1.$$

Use the bounded Cesàro test

$$\Psi(v) = \frac{1-(1-av)^K}{Ka} = v + \sum_{k=2}^{K} \frac{(-1)^{k+1}}{K}\binom{K}{k} a^{k-1} v^k. \tag{12}$$

It has zero constant coefficient, linear coefficient one, and $|\Psi(v_j)| \le 2/(Ka) = 1/(K\eta)$. By Lemma 7, its remaining averaged terms have modulus at most

$$\frac{2\sqrt{n+1}\,r}{Kan}\,\mathrm{e}^{-y+(2n+1)r}\left[\left(1+\frac{a}{r}\right)^K - 1 - \frac{Ka}{r}\right] \le B.$$

The triangle inequality and $m_1 \ge 1/2$ now imply

$$\frac{1}{2} \le m_1 \le \left|\frac{1}{n}\sum_j \Psi(v_j)\right| + B \le \frac{1}{K\eta} + B,$$

contrary to (11). Each individual residual is therefore larger than e^{-y}; taking its infimum gives the asserted non-strict bound. □

4 Choice of parameters

Proof of (2). Fix $C > C_0$, and choose $t > 0$ so small that

$$C^2 > \frac{4}{3}(2+t)\log 17.$$

Set

$$y = C\sqrt{n}, \qquad r = \frac{y}{8n}, \qquad \eta = \frac{y-\gamma_n}{n}, \qquad K = \left\lceil \frac{2+t}{\eta} \right\rceil.$$

For all sufficiently large n, the preliminary conditions of Proposition 8 hold: $\eta \sim C/\sqrt{n}$, $q \to 0$, and $\sqrt{n+1}\mathrm{e}^{-y} \to 0$. Also $K\eta \geq 2+t$, leaving a fixed positive margin in (11). We have

$$K \sim \frac{2+t}{C}\sqrt{n}, \qquad \frac{a}{r} \to 16,$$

and hence

$$\frac{-y + (2n+1)r + K\log(1+a/r)}{\sqrt{n}} \longrightarrow -\frac{3C}{4} + \frac{2+t}{C}\log 17 < 0.$$

The prefactor in (10) is $O(n^{-1})$. Thus $B \to 0$, and the finite certificate applies. □

Proof of (3). Take $n \geq 10^4$ and choose

$$y = 3\sqrt{n}, \qquad r = \frac{3}{8\sqrt{n}}, \qquad K = \left\lceil \frac{21}{10\eta} \right\rceil.$$

First,

$$\gamma_n \leq \frac{3}{20}\sqrt{n}, \qquad \frac{19}{20}\frac{y}{n} \leq \eta \leq \frac{y}{n}. \tag{13}$$

Indeed $\gamma_n/\sqrt{n}$ decreases for $n \geq 10^4$, since the sign of its derivative is that of $n/(n+1) - \gamma_n < 0$. At $n = 10^4$, $\gamma_n < 10 < 15$: $\mathrm{e}^8 > 2^8 = 256 > 2\sqrt{10001}$ suffices for the first inequality. In particular $0 < \eta < 2$. The condition $\sqrt{n+1}\mathrm{e}^{-y} < 1$ is immediate, and

$$q = \frac{3}{8}\sqrt{1+\frac{1}{n}}\exp\left(-\frac{9}{4}\sqrt{n} + \frac{3}{8\sqrt{n}}\right) < \frac{1}{2}.$$

Moreover $K\eta \geq 21/10$, so

$$\frac{1}{2} - \frac{1}{K\eta} \geq \frac{1}{42}.$$

From (13) we obtain $a/r \leq 16$ and $K \leq (14/19)\sqrt{n} + 1$. Using $\log 17 < 3$,

$$\begin{aligned} -y + (2n+1)r + K\log(1+a/r) &\leq \left(-\frac{9}{4} + \frac{14}{19}\log 17\right)\sqrt{n} + \log 17 + \frac{3}{8\sqrt{n}} \\ &< -\frac{3}{76}\sqrt{n} + 3 + \frac{3}{8\sqrt{n}} < 0. \end{aligned}$$

The last expression decreases with n and is negative at 10^4. For an elementary check of $\log 17 < 3$, the exponential series through degree five gives $\mathrm{e}^3 > 18 > 17$. Finally $Ka \geq 21/5$, and therefore

$$\frac{2\sqrt{n+1}\,r}{Kan} \leq \frac{5}{28n}\sqrt{1+\frac{1}{n}} < \frac{1}{5n}.$$

It follows that $B < 1/(5n) \leq 1/50000 < 1/42$. All conditions of Proposition 8 are satisfied. □

Remark 9. The constants 3 and 10^4 are convenient explicit choices. The same proof with $r = sy/n$, $0 < s < 1/2$, gives the sufficient asymptotic condition

$$C^2 > \frac{2\log(1+2/s)}{1-2s}.$$

We use $s = 1/8$ to state a simple constant. Neither this expression nor C_0 is asserted to be the optimal residual constant.

5 Exterior power sums and Hardy norms

Corollary 10. *Let $|z_j| \geq 1$, $1 \leq j \leq n$, and write $M_n = \max_{2\leq k\leq n+1} |\sum_j z_j^k|$. Then*

$$M_n \geq \frac{1}{n} \mathrm{e}^{-3\sqrt{n}} \qquad (n \geq 10^4).$$

For every $C > C_0$, uniformly over all such configurations, $M_n \geq n^{-1}\mathrm{e}^{-C\sqrt{n}}$ for all sufficiently large n. No distinguished point $z_1 = 1$ is needed.

Proof. We use the projected Newton identity of [1, Proposition 3.1]. Put $P(t) = \prod_{j=1}^n (1 - z_j t)$ and $p_k = \sum_j z_j^k$. Then $P \in \mathcal{S}_n$, $P(0) = 1$, and $P'(0) = -p_1$. Logarithmic differentiation near zero, followed by coefficient projection onto degrees at most n, gives exactly

$$P' - P'(0)P = -\Pi_{\leq n} \left[\left(\sum_{k=2}^{n+1} p_k t^{k-1} \right) P \right].$$

Projection is contractive in H^2, and multiplication by a polynomial S has H^2 norm at most $\|S\|_\infty$. Thus

$$s_n \leq \frac{\|P' - P'(0)P\|_2}{\|P\|_2} \leq \sum_{k=2}^{n+1} |p_k| \leq nM_n.$$

Apply Theorem 3. □

The factor $1/n$ is retained in these finite statements. It can be absorbed in a leading square-root exponent with any strictly larger constant. This corollary strengthens the scale in [1]; the earlier qualitative conclusion of [2] remains fully credited.

Corollary 11. *Uniformly for $1 \leq p \leq \infty$, $P \in \mathcal{S}_n$, and $\lambda \in \mathbb{C}$,*

$$\frac{\|P' - \lambda P\|_{H^p}}{\|P\|_{H^p}} \geq \frac{\mathrm{e}^{-3\sqrt{n}}}{\sqrt{n+1}} \qquad (n \geq 10^4).$$

For every $C > C_0$, the lower bound $\mathrm{e}^{-C\sqrt{n}}$ holds for all sufficiently large n, uniformly over the same parameters.

Proof. Write $R = P' - \lambda P$, of degree at most n. For $1 \leq p \leq 2$, bounding each Fourier coefficient by $\|R\|_1$ gives

$$\|R\|_2 \leq \sqrt{n+1}\|R\|_1 \leq \sqrt{n+1}\|R\|_p, \qquad \|P\|_2 \geq \|P\|_p.$$

For $p \geq 2$, use

$$\|R\|_2 \leq \|R\|_p, \qquad \|P\|_p \leq \|P\|_\infty \leq \sqrt{n+1}\|P\|_2.$$

In both cases $\|R\|_p/\|P\|_p \geq (\|R\|_2/\|P\|_2)/\sqrt{n+1}$. For the asymptotic statement, first apply Theorem 3 with a constant strictly between C_0 and C, and absorb $\frac{1}{2}\log(n+1) = o(\sqrt{n})$. □

6 Superalgebraic upper bounds

We prove the upper statement by cancelling a prescribed finite number of logarithmic-derivative terms at a strict global maximum. All constants in this section may depend on the prescribed cancellation order K. They are not asserted to be uniform as $K \to \infty$.

Lemma 12 (A finite positive moment cone)**.** *Fix $K \geq 3$. For a rational point $\zeta = x+iy$, $x > 0$, $y \geq 0$, define the real column*

$$V(\zeta) = \begin{cases} (\zeta^2, \dots, \zeta^K)^T, & y = 0, \\ (2\operatorname{Re}\zeta^2, \dots, 2\operatorname{Re}\zeta^K)^T, & y > 0. \end{cases}$$

There is a finite collection of these columns whose cone is $\mathbb{R}^{K-1}$. The collection can contain any prescribed K distinct positive rational real points. Its column matrix A_∞ admits a strictly positive rational vector α^0 satisfying

$$A_\infty \alpha^0 = (0, 3/2, 0, \dots, 0)^T.$$

Proof. A linear functional nonnegative on every such column would give a harmonic polynomial $h(z) = \operatorname{Re}\sum_{j=2}^{K} a_j z^j$ nonnegative on $\operatorname{Re} z > 0$, by density and conjugation symmetry. If its largest nonzero degree is $l \geq 2$, choose $\theta \in (-\pi/2, \pi/2)$ with $a_l \cos(l\theta) < 0$. Then $h(te^{i\theta}) < 0$ for large t, a contradiction. Thus the dual cone is zero and the closure of the generated cone is the whole space. Approximate each positive and negative coordinate vector by a finite conical combination closely enough that their convex hull still contains zero in its interior. The finite union of the columns used in these combinations already generates the whole space. Append the prescribed real points if necessary.

The resulting rational matrix has full row rank. For each column s_i, express $-s_i$ as a nonnegative combination of its columns. This representation can be rational: a nonempty rational polyhedron has a rational point. Adding these relations for all i gives a rational kernel vector with every coordinate strictly positive. A rational solution of the displayed full-rank system exists; adding a sufficiently large positive rational multiple of that kernel vector makes all of its coordinates positive. □

6.1 A fixed number of interior zeros

For a real reciprocal variable $v > 1$, put $\beta = 1 - 1/v$ and $f_v(z) = 1 - z/\beta$. For $v = x + iy$, $y > 0$, use the conjugate-pair factor $f_v(z) = (1 - z/\beta)(1 - z/\overline{\beta})$. Write $\nu_v = 1, 2$, respectively, and define

$$M_j(v) = \begin{cases} v^j, & v \text{ real}, \\ 2\operatorname{Re} v^j, & \Im v > 0, \end{cases} \qquad d(v) = \begin{cases} -v^2/(v-1), & v \text{ real}, \\ 2\operatorname{Re}[-v^2/(v-1)], & \Im v > 0. \end{cases}$$

Thus $d(v) = f_v'(0) - f_v'(1)/f_v(1)$, and $\operatorname{Re} v > 1/2$ is equivalent to $|\beta| < 1$. Use the circle-root baseline

$$B(z) = (1+z)^4(1 - z + z^2), \qquad b_j = \sum_{\beta:\, B(\beta)=0} (1-\beta)^{-j}.$$

It has degree six, constant coefficient one, $B'(0) = B'(1)/B(1) = 3$, $b_2 = 0$, and $b_3 = -3/2$.

Proof of Theorem 4. Fix $K \geq 3$. Apply Lemma 12, with K designated distinct positive real nodes among the ζ_i. Clear the denominators of its strictly positive weights by an integer Q, and put $m_i = Q\alpha_i^0$. These are positive integer group multiplicities, fixed independently of n. Set

$$M = \sum_i \nu_i m_i, \qquad S = \sum_i m_i M_1(\zeta_i) > 0.$$

The moments of orders $2, \dots, K$ are now $(0, 3Q/2, 0, \dots, 0)$. For every large integer n, define

$$R = (n/(6Q))^{1/3}, \quad \epsilon = R^{-1}, \quad c_n \equiv n - M \pmod 6, \quad |c_n - 2SR| \leq 3, \qquad b_n = (n - M - c_n)/6.$$

Choose a nearest integer in the specified congruence class; both b_n, c_n are positive eventually. Then $t_n = c_n/R = 2S + O(\epsilon)$ and $u_n = b_n/R^3 = Q + O(\epsilon^2)$.

Let ξ_i vary near ζ_i, varying only the K designated real nodes. Define $d_\epsilon(\xi) = -\xi^2/(\xi - \epsilon)$ for a real node, and twice its real part for a pair. Consider the K equations

$$0 = t/2 + \sum_i m_i d_\epsilon(\xi_i), \tag{14}$$

$$0 = t\epsilon/4 + \sum_i m_i M_2(\xi_i), \tag{15}$$

$$0 = b_j u\epsilon^{j-3} + t2^{-j}\epsilon^{j-1} + \sum_i m_i M_j(\xi_i), \qquad 3 \le j \le K. \tag{16}$$

They vanish at $(\epsilon, t, u, \xi) = (0, 2S, Q, \zeta)$. The Jacobian in the designated real positions has columns

$$m_i(-1, 2\zeta_i, 3\zeta_i^2, \dots, K\zeta_i^{K-1})^T.$$

It is a nonsingular scaled Vandermonde matrix. The real implicit-function theorem at this fixed limiting point gives $\xi_{i,n} = \zeta_i + O(\epsilon)$ solving the equations at (ϵ, t_n, u_n). Put $v_{i,n} = R\xi_{i,n}$ and

$$P_n(z) = B(z)^{b_n}(1+z)^{c_n}\prod_i f_{v_{i,n}}(z)^{m_i}.$$

For large n, its degree is exactly n, its constant coefficient is one, and its coefficients are real. Exactly M zeros, counted with multiplicity, lie strictly inside the disk; the other $n - M$ lie on the circle. The interior zeros satisfy $|1 - \beta_{i,n}| \asymp 1 - |\beta_{i,n}| \asymp R^{-1}$. Multiplying (14)–(16) by $R, R^2, \dots, R^K$, respectively, gives the exact identities

$$\lambda_n := P_n'(0) = P_n'(1)/P_n(1), \qquad \sum_{\beta:\, P_n(\beta)=0} (1-\beta)^{-j} = 0 \quad (2 \le j \le K).$$

In particular integer rounding has been corrected, and $\lambda_n = 3b_n + c_n/2 + R\sum_i m_i M_1(\xi_{i,n}) = O(n)$.

We give global estimates before using the cancellations locally. For $s = \sin^2(\theta/2)$,

$$|B(\mathrm{e}^{i\theta})/B(1)|^2 = \cos^6(\theta/2)\cos^2(3\theta/2) \le \mathrm{e}^{-3s}, \qquad |(1+\mathrm{e}^{i\theta})/2|^2 \le \mathrm{e}^{-s}.$$

For a real group $v = x > 1$, its normalized squared modulus is $1 + 4x(x-1)s \le \mathrm{e}^{T(v)s}$, with $T(v) = 4x(x-1)$. For a pair set $Y = y^2$, $U = x^2 + Y$, $V = (x-1)^2 + Y$. Its normalized squared modulus is

$$1 + 8(x^2 - x - Y)s + 16UVs^2 \le (1 + 4\sqrt{UV}s)^2 \le \mathrm{e}^{4(U+V)s}.$$

Indeed $UV - (x^2 - x - Y)^2 = (2x-1)^2 Y \ge 0$. Set $T(v) = 4(U+V)$ for this group. The fixed number of interior factors has total cost $\sum_i m_i T(v_{i,n}) = O(R^2)$. Since $R \asymp n^{1/3}$,

$$3b_n + c_n - \sum_i m_i T(v_{i,n}) = n/2 - O(R^2) \ge \kappa n$$

for some fixed $\kappa > 0$ and all sufficiently large n. It follows globally that

$$|P_n(\mathrm{e}^{i\theta})/P_n(1)|^2 \le \exp[-\kappa n \sin^2(\theta/2)]. \tag{17}$$

The zero-free neighborhood now shrinks with n. Choose a fixed $\delta > 0$ so that on $|z-1| \le \delta/R$ all $|v_{i,n}(z-1)| \le 1/2$, and the baseline zeros are excluded. Expanding $v/(1+v(z-1))$, the exact cancellations above give

$$|P_n'(z)/P_n(z) - \lambda_n| \le C_K(n + R^{K+1})|z-1|^K \le C_K R^{K+1}|z-1|^K. \tag{18}$$

The geometric-series remainder is uniform on this shrinking disk. There are $O(n)$ baseline reciprocal variables of bounded size and only M interior variables of size $O(R)$. By real symmetry the first angular derivative of the normalized log modulus vanishes at zero; its second derivative on a suitably smaller arc $|\theta| \le \delta'/R$ is $O(n + R^2) = O(n)$. Hence the normalized squared denominator is at least $cn^{-1/2}$, by integration over $|\theta| \le n^{-1/2}$. Combining (17) and (18), the local squared numerator is at most

$$C_K R^{2K+2} \int_{\mathbb{R}} \theta^{2K} \mathrm{e}^{-cn\theta^2}\, d\theta \le C_K R^{2K+2} n^{-K-1/2}.$$

Outside the small arc use the polynomial product derivative. Normalized derivatives of a baseline or extra factor are bounded; those of a real interior factor are $O(R)$, and those of a pair are $O(R^2)$. Removing a baseline factor reduces the positive global cost by three, removing an extra factor reduces it by one, and removing an interior group increases it by its positive $T(v)$. Thus every derivative term is bounded by a polynomial in n times the same Gaussian bound, up to a constant. On the outer arc $n\sin^2(\theta/2) \ge cn/R^2 \asymp n^{1/3}$. Its squared integral is therefore at most $\mathrm{poly}(n)\mathrm{e}^{-cn^{1/3}}$, negligible compared with every fixed inverse power. This estimate uses no rational quotient at a boundary zero. Dividing by the denominator proves

$$\rho(P_n)^2 \le C_K R^{2K+2} n^{-K} \le C_K n^{(2-K)/3}, \qquad \rho(P_n) \le C_K n^{(2-K)/6}.$$

Choose $K \ge 6A + 2$. Then $M_A = M$ is independent of n, as required. No explicit quantitative bounds are supplied here on M_A, C_A, or the starting degree as functions of A. □

Theorem 13 (One interior zero suffices)**.** *For every sufficiently large integer n, there is a real degree-n polynomial P_n, with $P_n(0) = 1$, exactly one zero strictly inside the unit disk and all remaining zeros on the circle, such that $\rho(P_n) = O(n^{-1/6})$. The interior zero is real and positive, with distance and radial gap from 1 of order $n^{-1/3}$. In conjunction with Theorem 5, the minimum number of interior zeros permitting a sequence with vanishing residual is exactly one. No optimal rate for one interior zero is asserted.*

Proof. We use two adjustable circle angles in the degree-34 baseline

$$B_{a,c}(z) = (1 - 2az + z^2)^{15}(1 - 2cz + z^2)^2, \qquad -1 < a, c < 1.$$

Its slope difference and second reciprocal moment are, respectively,

$$D(a, c) = -30a - 4c - 17, \qquad T(a, c) = -\frac{15a}{1-a} - \frac{2c}{1-c}.$$

Both vanish at $(a_*, c_*) = (-2/3, 3/4)$. Write $n - 1 = 34m + k$, $0 \le k \le 33$, and let $v = v_n > 3/2$ be the unique solution of

$$v^3 - \tfrac{3}{2}v^2 = (n-1)/4. \tag{19}$$

The left side is strictly increasing on this interval, and $v \sim (n/4)^{1/3}$. Set $\beta = 1 - 1/v \in (0, 1)$. For large n, $m > 0$. Define the real targets

$$d_n = \frac{v^2/(v-1) - k/2}{m} = O(n^{-2/3}), \qquad t_n = \frac{-v^2 - k/4}{m} = O(n^{-1/3}).$$

The following explicit formulas select circle parameters solving $D(a_n, c_n) = d_n$, $T(a_n, c_n) = t_n$:

$$\begin{aligned} L &= (51 + d_n)/2, \qquad U = 17 - t_n, \\ X &= \frac{UL + 221 + \sqrt{(UL - 289)(UL - 169)}}{30U}, \\ Y &= (L - 15X)/2, \qquad a_n = 1 - X, \qquad c_n = 1 - Y. \end{aligned}$$

Indeed these equations are equivalent to $15X + 2Y = L$, $15/X + 2/Y = U$; elimination gives $15UX^2 - (UL+221)X + 15L = 0$. The positive-square-root branch above tends to $X = 5/3, Y = 1/4$, since $UL \to 867/2$ and the square root tends to $391/2$. The denominators and discriminant stay positive, and $a_n, c_n \in (-1, 1)$ eventually, with deviations $O(n^{-1/3})$ from (a_*, c_*). These statements are uniform over the 34 values of k.

Put

$$P_n(z) = B_{a_n,c_n}(z)^m (1+z)^k (1 - z/\beta).$$

It has exactly the claimed degree, constant term and zero locations. The two target equations give the exact identities

$$P_n'(0) = P_n'(1)/P_n(1) =: \lambda_n, \qquad \sum_{j=1}^{n} v_j^2 = 0, \qquad v_j = (1-\beta_j)^{-1}.$$

Here the extra roots at -1 contribute $k/2$ to the slope difference and $k/4$ to the second moment; the interior root contributes $-v^2/(v-1)$ and v^2, respectively. For a conjugation-symmetric multiset of N circle roots avoiding 1, its reciprocal moments satisfy

$$M_3^\circ = \tfrac{3}{2} M_2^\circ - N/4. \tag{20}$$

This follows for each pair $1/2 \pm iy$, and also for the real reciprocal $1/2$. Conjugation symmetry is essential and holds here. Taking $N = n-1$ and $M_2^\circ = -v^2$, (19) gives $\sum_j v_j^3 = 0$ exactly. Moreover $\lambda_n = (n-1)/2 + v = O(n)$.

We verify the global norm control for this new baseline. In $s = \sin^2(\theta/2)$, its limiting normalized squared modulus is

$$H(s) = (1 - 6s/5)^{30}(1-8s)^4, \qquad 0 \le s \le 1.$$

Apart from its zeros $1/8, 5/6$, the only interior stationary point is $5/24$. Its values there and at 1 are

$$H(5/24) = (3/4)^{30}(2/3)^4 < 1, \qquad H(1) = 7^4/5^{30} < 1.$$

Since $H(0) = 1$, zero is the unique global maximum. Also $\log H(s) = -68s + O(s^2)$. The nearby circle parameters retain uniform negative curvature near zero and, by compactness, a strict modulus gap away from zero. Therefore, for some fixed $\kappa > 0$ and all sufficiently large n,

$$|B_{a_n,c_n}(\mathrm{e}^{i\theta})/B_{a_n,c_n}(1)|^2 \le \mathrm{e}^{-\kappa s} \qquad (-\pi \le \theta \le \pi).$$

The interior factor contributes $1 + 4v(v-1)s \le \exp(4v(v-1)s)$, whereas the -1 factors contribute at most e^{-ks}. Since $m \sim n/34$ and $v^2 = O(n^{2/3})$, $\kappa m + k - 4v(v-1) \ge cn$ eventually. Thus

$$|P_n(\mathrm{e}^{i\theta})/P_n(1)|^2 \le \exp[-cn \sin^2(\theta/2)] \tag{21}$$

globally, for a fixed $c > 0$.

The circle reciprocals are uniformly bounded and only the interior reciprocal is large. On $|z-1| \le \delta/v$, for a fixed small $\delta > 0$, all denominators below stay separated from zero. The exact slope and moment cancellations imply

$$F_n(z) := P_n'(z)/P_n(z) - \lambda_n = -(z-1)^3 \sum_{j=1}^{n} \frac{v_j^4}{1 + v_j(z-1)}.$$

Consequently $|F_n(z)| \le C(n+v^4)|z-1|^3 \le Cn^{4/3}|z-1|^3$. On a smaller angular window of radius δ'/v, real symmetry makes the first derivative of the normalized log modulus zero at zero, and its second derivative is $O(n+v^2) = O(n)$. As $n^{-1/2} = o(v^{-1})$, integration on $|\theta| \le n^{-1/2}$

bounds the normalized squared denominator below by $c'n^{-1/2}$. The local squared numerator is at most

$$Cn^{8/3}\int_{\mathbb{R}}\theta^6 \mathrm{e}^{-c'n\theta^2}\,d\theta = O(n^{-5/6}).$$

On the outer arc differentiate the polynomial product, avoiding logarithmic-derivative poles. The normalized derivative of a baseline factor is uniformly bounded, its multiplicity is $O(n)$, and the normalized interior derivative is v. Removing a baseline loses only κ in the global exponent; removing the interior factor improves it. The extra-factor multiplicity is bounded. The term $-\lambda_n P_n$ costs at most another $O(n)$. Thus the outer squared integral is bounded by a polynomial in n times $\exp(-c'n/v^2) = \exp(-c''n^{1/3})$. Dividing both numerator estimates by the same denominator gives $\rho(P_n)^2 = O(n^{-1/3})$, as required.

Finally, Theorem 5 excludes vanishing residuals with zero interior roots; this construction permits them with one. This proves the zero-count threshold, not a sharp rate or the superalgebraic conclusion with one root. No explicit starting degree is claimed. The next proposition identifies the asymptotic norm constant of this particular family. □

Proposition 14 (Asymptotic constant for the explicit family)**.** *For the polynomials constructed in Theorem 13, through all sufficiently large integer degrees,*

$$\rho(P_n) \sim \frac{\sqrt{15}}{4^{4/3}} n^{-1/6}.$$

This constant describes that family, not a universal one-root optimum.

Proof. Retain the construction's notation and put $G_n = P_n/P_n(1)$, $F_n = P_n'/P_n - \lambda_n$, and $w = z-1$. Its reciprocal zeros satisfy $S_1 = \lambda_n = (n-1)/2 + v$, $S_2 = S_3 = 0$, and $v/n^{1/3} \to 4^{-1/3}$. All boundary reciprocal zeros are uniformly bounded. The exact finite geometric remainder is

$$F_n(z) = -w^3 \sum_j \frac{v_j^4}{1+v_j w}.$$

At $z = \exp(iu/\sqrt{n})$, for each fixed real u, the boundary contribution to the sum is $O(n)$, whereas the single interior contribution is asymptotic to v^4. Hence

$$n^{1/6} F_n(\mathrm{e}^{iu/\sqrt{n}}) \longrightarrow i4^{-4/3}u^3. \tag{22}$$

The local logarithm, normalized to vanish at 1, has expansion

$$\log G_n(1+w) = \lambda_n w + O((n+v^4)|w|^4),$$

by the exact second- and third-moment cancellations. For fixed u the remainder tends to zero, and $\lambda_n/n \to 1/2$, giving

$$|G_n(\mathrm{e}^{iu/\sqrt{n}})|^2 \longrightarrow \mathrm{e}^{-u^2/2}. \tag{23}$$

The preceding proof gives the global Gaussian bound $|G_n(\mathrm{e}^{it})|^2 \le \exp[-cn\sin^2(t/2)]$, and locally $|F_n(z)| \le Cn^{4/3}|z-1|^3$ for $|z-1| \le \delta/v$. In the rescaled variable $u = t\sqrt{n}$, the denominator is dominated by $\exp(-c'u^2)$. On the local region the rescaled numerator $n^{1/3}|G_nF_n|^2$ is dominated by $C|u|^6\exp(-c'u^2)$; that region expands to the whole line at rate $n^{1/6}$. Outside it, the product-derivative tail estimate from the same proof is polynomial in n times $\exp(-cn^{1/3})$, negligible even after multiplication by $n^{5/6}$. This use of the polynomial product avoids a logarithmic quotient at a boundary zero. Dominated convergence in (22)–(23) therefore yields

$$\sqrt{n}\|G_n\|_2^2 \longrightarrow \frac{1}{\sqrt{2\pi}}, \qquad n^{5/6}\|G_nF_n\|_2^2 \longrightarrow \frac{15\,4^{-8/3}}{\sqrt{2\pi}},$$

where 15 is the sixth standard Gaussian moment. Dividing proves the claim. All constants in the preceding construction are uniform over the finite set of 34 degree remainders, so the limit holds through all sufficiently large integers. □

7 Interior-root budgets and their obstruction

We first prove the all-degree bound and then the existence criterion. The proof also identifies the structural role of the circle roots.

7.1 Multiplication by a polynomial of small degree

For $T = c\prod_j (z-\gamma_j)$, write $\mathcal{M}(T) = |c|\prod_j \max(1,|\gamma_j|)$ for its Mahler measure. The classical coefficient bound gives

$$\|T\|_2 \le \sqrt{\binom{2\ell}{\ell}}\,\mathcal{M}(T) \le 2^\ell \mathcal{M}(T), \qquad \ell = \deg T. \tag{24}$$

Indeed each coefficient is bounded by the corresponding binomial coefficient times $\mathcal{M}(T)$, by expanding the product; sum their squares. See also [8, Lemma 1.6 and Exercise 1.11].

Lemma 15. *If P has degree n and $H \neq 0$ has degree at most d, then*

$$\|HP\|_2 \ge \|H\|_2 (n+d+2)^{-d}\|P\|_2.$$

Proof. For $V = \sum_{j=0}^N a_j z^j$ and $\gamma \in \mathbb{C}$, Parseval gives

$$\|(z-\gamma)V\|_2^2 = (1+|\gamma|^2)\|V\|_2^2 - 2\,\mathrm{Re}\left(\overline{\gamma}\sum_{j=0}^{N-1} a_j\overline{a_{j+1}}\right).$$

The numerical radius of the truncated shift of dimension $N+1$ is $c_N = \cos(\pi/(N+2))$; this is the classical extremal shift calculation in [9]. Directly, after a diagonal phase rotation its real part is the path matrix with off-diagonal entries $1/2$. Its eigenvectors have coordinates $\sin((j+1)k\pi/(N+2))$, and its largest eigenvalue is c_N. Thus the squared norm ratio is at least $1+r^2-2rc_N$, where $r=|\gamma|$. For $r \le 1$, completing a square bounds this below by $1-c_N^2$; for $r \ge 1$, divide by r^2 and use the same argument at $1/r$. Consequently

$$\|(z-\gamma)V\|_2 \ge \max(1,|\gamma|)\sin\frac{\pi}{N+2}\|V\|_2.$$

Factor H into $\ell \le d$ linear factors and apply this estimate successively at degrees $n, n+1, \dots, n+\ell-1$. Since $\sin(\pi/x) \ge 2/x$ for $x \ge 2$,

$$\|HP\|_2 \ge \mathcal{M}(H)\left(\frac{2}{n+d+2}\right)^\ell \|P\|_2 \ge \|H\|_2(n+d+2)^{-\ell}\|P\|_2.$$

This implies the assertion; the constant-polynomial case is immediate. □

7.2 A positive semicircle in the reciprocal identity

Proof of Theorem 1. Write $P = cAQ$, where $A(z) = \prod_{j=1}^M (z-\beta_j)$, $|\beta_j| < 1$, is monic and all roots of Q lie on the circle. For a polynomial considered with degree parameter k, set $T^*(z) = z^k\overline{T(1/\bar z)}$. Use parameters n for P, M for A, and importantly n for $R = P' - \lambda P$, even if R has smaller actual degree. Self-inversivity of Q gives $P/P^* = \omega A/A^*$ for some $|\omega| = 1$. The identity $z^2(P^*)' = nzP^* - R^* - \bar\lambda P^*$, followed by differentiation of this quotient and clearing denominators, yields

$$z^2 AA^*R + \omega A^2 R^* = HP, \tag{25}$$

$$H = z^2\{A'A^* - A(A^*)'\} + (nz - \lambda z^2 - \bar\lambda)AA^*. \tag{26}$$

These are polynomial identities, including at zeros of the factors. Here $\deg H \le d = 2M+2$, and on the circle $|A^*| = |A| \le 2^M$, $|R^*| = |R|$. Hence

$$\|HP\|_2 \le 2\,4^M \|R\|_2. \tag{27}$$

We obtain a lower bound for H without a separation assumption. The finite Blaschke identity, also given in [11, Lemma 4.3], is

$$z\frac{(A/A^*)'}{A/A^*} = \sum_{j=1}^{M} \frac{1-|\beta_j|^2}{|z-\beta_j|^2}, \qquad |z| = 1.$$

It follows either by differentiating each factor or by its Poisson kernel formula. Since $AA^* = z^M|A|^2$ on the circle,

$$z^{-M-1}H(z) = (n - 2\,\mathrm{Re}(\lambda z))|A(z)|^2 + K_A(z), \tag{28}$$
$$K_A(z) = \sum_{j=1}^{M}(1-|\beta_j|^2)\prod_{\ell\neq j}|z-\beta_\ell|^2 \ge 0.$$

Choose a semicircle E on which $\mathrm{Re}(\lambda z) \le 0$; if $\lambda = 0$, choose either semicircle. Thus $|H| \ge n|A|^2$ on E.

Jensen's formula for the monic stable polynomial A gives $\int_{\mathbb{T}} \log|A|\,dt/(2\pi) = 0$. On the complementary semicircle, $\log|A| \le M\log 2$. Therefore its average over E, with measure dt/π, is at least $-M\log 2$. Applying Jensen's inequality to $\exp(4\log|A|)$ on E gives

$$\frac{1}{\pi}\int_E |A|^4\,dt \ge 2^{-4M}, \qquad \|H\|_2 \ge \frac{n}{\sqrt{2}\,4^M}.$$

This restricted-arc Jensen argument is classical in origin; compare [10, Lemma 4.1]. The logarithm is continuous on the circle for each strictly interior configuration, so no limiting interchange as the roots approach the boundary is needed. In particular $H \neq 0$. Lemma 15 and (27) now give (1). For $n \ge 2M+4$, use $n+2M+4 \le 2n$ and weaken $2^{-6M-7/2}$ to 2^{-6M-4}. □

Corollary 16 (Necessary quantitative count)**.** *Under the hypotheses of Theorem 1,*

$$M \ge \frac{\log(1/\rho_\lambda(P)) - \log n - \log(98\sqrt{2})}{2\log n + \log 784}.$$

In particular, a sequence with $\rho_{\lambda_n}(P_n) \le \exp(-cn^\alpha)$, where $c, \alpha > 0$, necessarily has $M_n \ge (c/2 + o(1))n^\alpha/\log n$.

Proof. Since $M \le n$, we have $n + 2M + 4 \le 7n$. Taking logarithms in (1) gives

$$\log(1/\rho_\lambda(P)) \le (2M+1)\log n + M\log 784 + \log(98\sqrt{2}).$$

Rearrangement proves the result. This is only a necessary condition; Theorem 3 can also exclude a proposed decay rate. □

Remark 17 (Stable factor and self-inversive cofactor). The same bound (1) holds for any degree-n $P = cAQ$ with A monic of degree M, all its zeros strictly inside the disk, and $Q = \eta Q^*$, $|\eta| = 1$. Indeed only this last identity was used for Q; the remaining estimates use the degree of P and the zeros of A. The cofactor may therefore contain conjugate-reciprocal pairs inside and outside the circle. In this extension M means the degree of the chosen stable factor, not the total number of interior zeros of P. Not every polynomial has such a factorization. For instance, $P_R = 1 - z/R$, $R > 1$, with $\lambda = -1/R$ has residual $1/[R^2\sqrt{1+R^{-2}}] \to 0$ at fixed degree one; its unpaired exterior root falls outside the hypothesis.

7.3 Sufficiency, necessity and exact saturation

Proof of Theorem 2. If b_n does not tend to infinity, there are a finite B and infinitely many degrees with $b_n \le B$. For every such sufficiently large n, Theorem 1 bounds any admissible residual below by $2^{-6B-4}n^{-2B-1}$, since its actual count $M \le B$. This contradicts superalgebraic decay, so (2) implies (1).

Suppose now $b_n \to \infty$. For each integer $j \ge 1$, apply Theorem 4 with exponent $2j$, obtaining M_j, C_j, N_j. Choose strictly increasing integers $T_j \ge j$ so large that for every $n \ge T_j$,

$$b_n \ge M_j, \qquad n \ge N_j, \qquad n^j \ge C_j.$$

All auxiliary eventual conditions in that construction can also be included in N_j. With $j(n) = \max\{j : T_j \le n\}$, the constructed real normalized polynomial has exactly $M_{j(n)}$ interior zeros and restricted residual at most $n^{-j(n)}$. The index is finite and tends to infinity.

To achieve exactly b_n interior zeros while retaining reality, use the explicit boundary factor in Section 6:

$$B_6(z)^u(1+z)^v, \qquad B_6(z) = (1+z)^4(1-z+z^2), \quad u, v \ge 0.$$

It supplies $4u+v$ real roots at -1 and u conjugate pairs, totaling $6u+v = n - M_{j(n)}$ roots. Eventually $u \ge 1$. Given the required additional count $s = b_n - M_{j(n)}$, take

$$q = \min\{u, \lfloor s/2 \rfloor\}, \qquad r = s - 2q.$$

Then $0 \le q \le u$ and $0 \le r \le 4u+v$. Move q conjugate pairs and r real roots radially inward by a common sufficiently small positive amount. This preserves real coefficients, the degree and nonzero constant term. After normalizing at zero, the coefficients and the restricted slope tend to their original values as the displacement tends to zero. The restricted residual is continuous because its norm denominator is nonzero. For each n, choose the displacement so that the residual increases by at most $n^{-j(n)}$. The final polynomial has exactly b_n interior zeros and residual at most $2n^{-j(n)} = o(n^{-A})$ for every fixed $A > 0$. The finitely many initial degrees can be filled arbitrarily with real normalized polynomials having the prescribed count. This proves (1) implies (3), while (3) implies (2) immediately.

For residuals merely tending to zero, infinitely many zero budgets would contradict Theorem 5. Conversely, a budget at least one eventually accommodates the family of Theorem 13. This last assertion concerns an upper budget, not saturation of every budget by that particular family. □

For a fixed M, residual $O(n^{-A})$ consequently requires $A \le 2M+1$. This necessary relation is not a matching tradeoff. The number M_A in Theorem 4 depends on A; no single fixed count gives a superalgebraic sequence.

7.4 A stronger obstruction for one interior zero

The reciprocal identity works for every count. At count one, the third and fourth reciprocal moments give a stronger exponent.

Proposition 18. *If $n \ge 4096$, $P \in \mathcal{S}_n$ has exactly one strictly interior zero and all other zeros on the circle, then, for all $\lambda \in \mathbb{C}$,*

$$\rho_\lambda(P) \ge \frac{n^{-13/6}}{512\sqrt{2}\,\mathrm{e}^3}.$$

Proof. Set $C = 2\sqrt{2}\,\mathrm{e}^3$ and suppose for contradiction that $\rho = \rho_\lambda(P) \le n^{-13/6}/(256C)$. Rotate a maximum point to 1 and normalize $G(1) = \|G\|_\infty = 1$ as in Lemma 6. Its initial part gives $|\mu| \le n+1$ and $S_1 = G'(1) \in \mathbb{R}$, where $S_k = \sum_j (1-\beta_j)^{-k}$ uses the rotated zeros. The local

ODE proof of Lemma 7, at radius $1/n$, does not require its separate separation parameter η. Here its smallness parameter is at most $\mathrm{e}^3/n < 1/2$, and it yields the unaveraged estimates

$$|S_k| \le 2\sqrt{n+1}\rho \mathrm{e}^{(2n+1)/n} n^{k-1} \le C\rho n^{k-1/2}, \qquad k \ge 2.$$

In particular,

$$|S_2| \le \frac{n^{-2/3}}{256}, \quad |S_3| \le \frac{n^{1/3}}{256}, \quad |S_4| \le \frac{n^{4/3}}{256}. \tag{29}$$

The $n-1$ boundary reciprocals are $w_j = 1/2 + it_j$, $t_j \in \mathbb{R}$; the remaining reciprocal is $v = x+iy$, $x > 1/2$. Since S_1 is real, $\sum_j t_j = -y$, so

$$\operatorname{Im} S_2 = (2x-1)y,$$

and, using $\operatorname{Re} w_j^3 = (3/2)\operatorname{Re} w_j^2 - 1/4$,

$$x^3 - \tfrac{3}{2}x^2 - 3(x-\tfrac{1}{2})y^2 = \frac{n-1}{4} + \operatorname{Re} S_3 - \tfrac{3}{2}\operatorname{Re} S_2 \ge \frac{n}{8}.$$

The last inequality follows from (29) for $n \ge 4096$. Both subtracted terms on the left are nonnegative, so $x \ge n^{1/3}/2 \ge 8$. Consequently $2x - 1 \ge x$, $|xy| \le |S_2|$, and

$$\operatorname{Re} v^4 = x^4 - 6x^2y^2 + y^4 \ge \frac{n^{4/3}}{16} - \frac{6}{65536} n^{-4/3}.$$

For each boundary reciprocal, $\operatorname{Re}(1/2+it)^4 = (t^2 - 3/4)^2 - 1/2 \ge -1/2$. It follows that

$$\operatorname{Re} S_4 \ge \frac{n^{4/3}}{16} - \frac{6}{65536} n^{-4/3} - \frac{n-1}{2} \ge \frac{n^{4/3}}{64},$$

using $n/2 \le n^{4/3}/32$. This contradicts (29). □

Together with Theorem 13, this leaves a gap between the universal one-root lower exponent 13/6 and the achieved upper exponent 1/6. No optimality of either is asserted.

8 A fixed radial gap forces a polynomial barrier

The upper construction uses the full closed disk. A fixed strict root radius gives a complementary obstruction, by the same local estimate and bounded test as the lower-bound proof.

Proposition 19. *For $0 \le q < 1$, let $s_n(q)$ denote the infimum defining s_n with all zeros restricted to $|z| \le q$. Set*

$$a = 1 - q^2, \quad K = \lceil 4/a \rceil, \quad \Delta = \frac{1}{1+q} - \frac{1+q^K}{Ka} > 0, \qquad c_q = \min\left\{ \frac{1}{\sqrt{2}}, \frac{\Delta K a}{2\sqrt{2}\mathrm{e}^3 (1+a)^K} \right\}.$$

Then $s_n(q) \ge c_q n^{3/2-K}$ for every $n \ge 41$.

Proof. Write ρ for an admissible residual. If $\rho > 1/\sqrt{n+1}$, the conclusion is immediate. Otherwise the maximum normalization gives $|\mu| \le n+1$. Take $r = 1/n$. The smallness parameter in the proof of Lemma 7 is at most $\mathrm{e}^3/n < 1/2$. That local proof does not need the separate zero-separation parameter η, and gives

$$|m_j| \le \frac{2\sqrt{n+1}}{n} \rho \mathrm{e}^3 r^{1-j} \qquad (j \ge 2).$$

The exact image of $|\beta| \le q$ under $v = (1-\beta)^{-1}$ is $|v - 1/a| \le q/a$. Thus $|1 - av| \le q$ and $\operatorname{Re} v \ge 1/(1+q)$. Use $\Psi(v) = [1-(1-av)^K]/(Ka)$, whose modulus is at most $(1+q^K)/(Ka)$. Estimating its higher averaged terms gives

$$\Delta \le \frac{2\mathrm{e}^3\sqrt{n+1}\rho}{Kan^2}[(1+an)^K - 1 - Kan] \le \frac{2\mathrm{e}^3\sqrt{n+1}\rho}{Kan^2}(1+an)^K.$$

Now $\sqrt{n+1} \le \sqrt{2n}$ and $1+an \le (1+a)n$. The resulting lower bound is the stated one. Finally $Ka \ge 4$, $q^K < 1$, and $1/(1+q) > 1/2$ ensure $\Delta > 0$. □

In particular, any single sequence of admissible degree-n polynomials with residual $o(n^{-A})$ for every fixed $A > 0$ must have its maximum root modulus tending to one. Otherwise an infinite subsequence would satisfy a fixed polynomial lower bound. No sharp radial exponent is asserted.

9 Scope and verification

The uniform slope formulation bounds approximate eigenvectors of polynomial differentiation under a zero-location constraint. It does not describe the full unconstrained pseudospectrum. For example, for $u = n/4$ and $T_n(z) = \sum_{k=0}^n (uz)^k/k!$,

$$T_n' - uT_n = -\frac{u^{n+1}}{n!}z^n, \qquad \frac{\|T_n' - uT_n\|_2}{\|T_n\|_2} \le \frac{n}{4}\left(\frac{\mathrm{e}}{4}\right)^n,$$

using $\|T_n\|_2 \ge 1$ and $n! \ge (n/\mathrm{e})^n$. Thus a root constraint is essential for a bound of our form.

The lower and upper disk bounds do not match in order. Whether the restricted exterior power-sum minima stay bounded away from zero remains unresolved here, as does the optimal order of s_n and r_n. The superalgebraic upper bound is not a stretched-exponential upper bound, and its constants are not optimized. The minimal interior-root count permitting collapse is one, but the optimal rate with one such root is not established. Theorem 1 excludes superalgebraic decay for every fixed number of interior roots. The elementary radial barrier is a consequence of the same local method. We do not assert a theorem for arbitrary starting exponents, weighted power sums, or H^p with $p < 1$. The elementary norm comparison in Corollary 11 is a consequence of the main estimate.

The analytic proof of the explicit threshold uses only the displayed elementary inequalities. It does not rely on optimization or sampling. The supplementary diagnostics check exact projected Newton identities over Gaussian rationals, the bounded test on admissible reciprocal zeros, binomial identities, and numerical instances of the finite certificate. Rational checks of constants and high-precision evaluations are included for reproducibility. The circle-root family is checked with complete exact coefficient Parseval ratios. The one-interior-root family has symbolic and high-precision checks of the cubic, slope, moments and circle parameters. A separate calculation by the same involved model reviewer covers all 34 degree classes at two scales. Additional exact checks cover the reciprocal identity, its complex conjugations, the positive Poisson kernel, the one-root moment obstruction and real-root saturation.

For Proposition 14, supplementary numerical norm diagnostics use two equispaced grids, of sizes $n+1$ and $n+2$, at eight degrees up to 10 000 017. Such grids would integrate the squared polynomial norms exactly in exact arithmetic. The implemented evaluations instead use rounded parameters and omit underflow-scale weights. The cancellation-free remainder approximates the exact family; it is not literally the exact residual of the polynomial with all parameters rounded. Agreement between grids does not certify the common rounding or masking error. These computations are not interval-certified norm evaluations and do not prove the asymptotic limit. Finite moment searches motivated the general construction but are not premises of its

arbitrary-order proof. None of the checks establishes historical priority or replaces the analytic arguments.

The comparison uses the exact primary-source versions listed below. A targeted search did not identify the bounds in this exact setting; exhaustive priority certification is not claimed. The accessible Komarov versions listed below were used for the stated comparison. A related 2020 journal text (DOI 10.1070/IM8901) was not available in full in this review; no exclusion of every possible bibliographic overlap is claimed. AI tools assisted exploration, drafting and computational checking. Model review is distinct from independent human peer review and formal proof verification.

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